

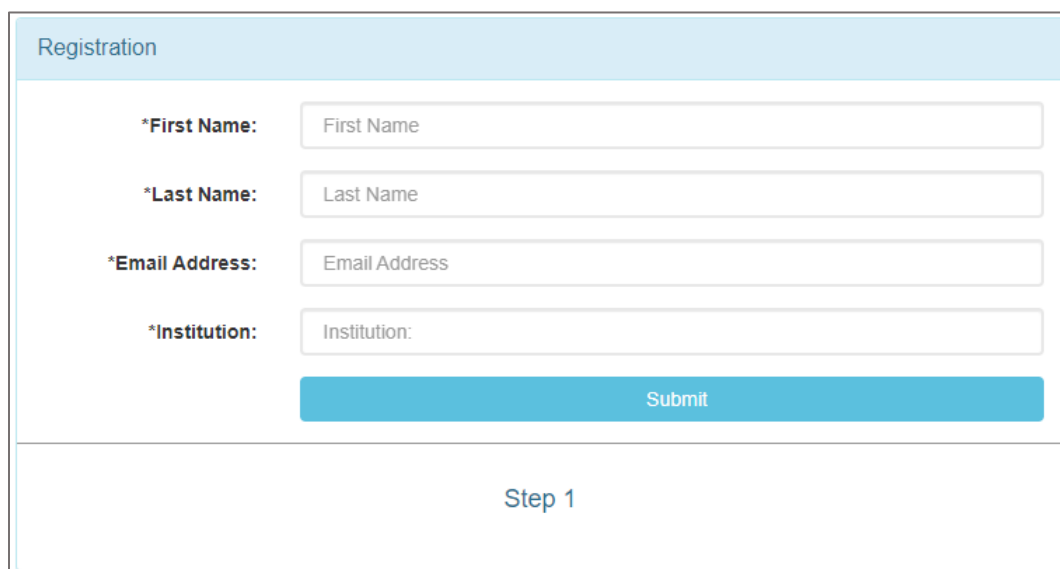
Documentation Available on the Verasonics Community Portal

In an effort to minimize the size of the Verasonics Software Release package, the Vantage™ product documentation is made available to users on the [Verasonics Community Web Portal](https://verasonicscommunity.com/). Access to this site is restricted to Verasonics customers and requires a one-time registration.

To access documentation on the Verasonics Community Web Portal, complete the registration process by following the steps below:

1. Navigate to the Verasonics Community Portal using the URL - <https://verasonicscommunity.com/>.
2. Click the *Register* button.
3. On the registration page, fill in the required information and click the *Submit* button.

Note: Please enter your institutional email address only.



The image shows a web form titled "Registration" in a light blue header. Below the header, there are four labeled input fields: "*First Name:" with a placeholder "First Name", "*Last Name:" with a placeholder "Last Name", "*Email Address:" with a placeholder "Email Address", and "*Institution:" with a placeholder "Institution:". Below these fields is a blue "Submit" button. At the bottom of the form, there is a light blue box containing the text "Step 1".

Figure 1: Registration Page

4. On the next page, provide your title and biodata. It is useful to provide this information if you intend to contribute scripts to the broader community on this portal—it enables other users to see all contributions associated with your account.
5. After you enter a password of your choice, an authorization code will be sent to your email.
Note: If you do not see an email (with the authorization code) in your inbox, check your junk mail or other spam mail folders.
6. On the next page, enter the authorization code sent to your email.
7. After entering the code, re-enter your password and agree to the terms of use by scrolling down and checking the 'I agree' field. Then click the *Submit* button.

Note: Please reach out to the Verasonics community administrator at CommunityAdmin@Verasonics.com if you have any questions about the registration process.

After obtaining access to the Verasonics Community Web Portal, click on the **Vantage Resource Documents** tab to view the Verasonics product documentation.

You can view a list of application notes, user manuals, a script repository, and other helpful documents here. You have the option to narrow down the list of documents by entering a keyword in the *Search* box.

Application Note: Wall Filter Frequency Responses Under CDI Weak/Low Flow Conditions

This document describes the performance of weak/low flow wall filter coefficient sets available to the Doppler Process Object defined in the Sequence Programming Manual.

Application Note: Volume Imaging Package

The document introduces the user to the Multi-System Volume Imaging Vantage product and provides details on how to properly write a script to program and operate the multi-system configuration. The document provides an overview, hardware setup, sequence programming details, remote commands, initiating a sequence, and a list of known bugs and issues.

Application Note: Vantage Example Scripts - NDE/NDT

This document provides a matrix summary of all available example scripts in the directory: Examples_NDE_NDT.

Application Note: Vantage Analog Signal Path Gain

This document provides a summary of the analog signal path gain of the Vantage system.

Application Note: Use of HVMUX Transducer with Vantage System

This document describes user-programmable features included in the Vantage system that enable power and control for HVMUX switches located within a transducer.

Application Note: New UI Scheme

This document outlines a modern object-oriented design for defining UI controls. The new design brings together the flexibility and familiarity of defining sequences, with new features that make script development easier.

Application Note: ShowTXPD Tool

The showTXPD tool enables users to visualize the alignment of the transmit beam with the pixel data. This document describes the features of the showTXPD tool.

Application Note: Shear Wave Visualization Script - Description and Operation

This document describes how to configure and use the shear wave visualization script. Important considerations prior to the use of this script are also provided.

Application Note: Quadrature Sampling

This document provides analysis and illustrations to show the generation of true analytic signals with direct quadrature sampling.

Application Note: Process Storage Class - Data Import and Export

This document outlines the functionality of the process storage class, which provides an efficient method for defining sequence-programming-based RF and image-data file import and export. In addition, it describes parameters that are supported by the process storage class.

Application Note: Probe Temperature Sensing

The Vantage system provides hardware circuitry and software utilities to allow monitoring of one or two transducer temperature sense thermistors connected through dedicated pins at the probe connector, for probes that provide this feature. This document describes the functionality of the hardware feature, how to program it for monitoring and for enabling the automatic shutdown threshold, and how to convert the monitored value into a temperature.

Application Note: Phase-Accurate Synchronization of Multiple Vantage Systems Using the Multi-System Synchronization Module

This document describes the use of the Synchronization Module hardware and software configuration needed to share a single 250 MHz system clock between multiple Vantage systems, and thus enable synchronous and phase-aligned operation of multiple hardware sequencers. This arrangement permits coherent acquisitions with up to 2048 channels (using up to 8 Vantage 256 systems; it is also possible to synchronize different Vantage models). Each Vantage system operates with its own host controller when running by itself, and the data collected by each system can be processed as is normally done for real-time display. The data can also be transferred to a common host controller during the sequence or after the experiment and processed coherently using the complete data set.

Application Note: NDE Research Software

The document introduces users to the NDE Research Software and provides instructions for its use. The features of the software are described, and each user setting is explained. In addition, instructions and examples are also provided for the development of custom imaging modes.

Application Note: Multithreading Framework

The document outlines the implementation of a multithreading framework for parallelizing code and fully leveraging the benefits offered by multicore and multi-CPU systems.

Application Note: Introduction to Waveform Generation Applications in Ultrasound

This document introduces waveform generation applications in ultrasound and provides an overview of conventional transmit waveforms and coded excitations.

Application Note: Vantage HIFU Configuration Setup Instructions

This document provides instructions on the wiring connections, initial setup, and functionality testing of the AIM-TTI QPX600DP external transmit power supply for use with a Vantage system configured with the HIFU transmit option.

Application Note: Extended Input/Output

The document introduces users to the Extended I/O option functionality and typical use cases. Detailed instructions are provided for the installation and programming of the option, including the use of setup objects and process functions. To aid understanding, this document is best used in conjunction with the provided example scripts.

Application Note: Event Analysis Tool

This document provides an overview of the Event Analysis Tool that can assist with developing and debugging a setup script. In addition, it offers debugging strategies for identifying errors in the script and describes the capabilities of the tool that can be used for debugging setup scripts.

Application Note: DC Bias for Transducer Element

This document summarizes how to apply DC bias voltage to transducer elements while using a Vantage hardware system.

Application Note: Cannon Connector to Cable Interface PCB - Signal Breakout Board

This document specifies the requirements and design for an interface PCB assembly and associated shields to allow convenient connection of cables to the transducer connector used on the Vantage hardware system.

Application Note: Arbitrary Waveform Generation Toolbox

This document describes the use of the Verasonics Arbitrary Waveform Generation Toolbox, called the GUI-based arbitrary waveform generation tool (GUI-AWE), version v1.

Arbitrary Waveforms using a Tri-State Transmit Pulser

The paper outlines methods for encoding waveforms into sequences suitable for a tri-state RF ultrasonic transmitter, under various fidelity criteria.

Application Note: Use Models and Fidelity Criteria for the Verasonics Arbitrary Waveform Encoders

This document explains the use models and fidelity metric concepts needed for users of the Verasonics Arbitrary Waveform (AW) Encoder tools.

Application Note: UTA 1024-MUX Adapter

This document describes the intended use of the Vantage UTA 1024-MUX adapter, the required configuration, programming model, assembly, and the basic sequence needed to complete a 1024-element acquisition made up of multiple transmit-receive events.

Tutorial for Classifying Objects in Ultrasound Data

This tutorial is designed to be used as a guide on how to use the MATLAB® Statistics and Machine Learning toolbox in combination with the Verasonics Vantage system for live classification. It contains four easy steps that allow users to understand the overall classification workflow. The parameters in this tutorial work best for classifying the large hyperechoic circles in images from a CIRS Multi-Purpose Multi-Tissue Ultrasound Phantom Model 040GSE. This tutorial requires the use of an L11-5v transducer, MATLAB's Statistics and Machine Learning Toolbox, and MATLAB's Image Processing Toolbox.

FUS Elite User Manual

The FUS Elite User Manual contains important safety and regulatory guidelines for proper use of the system, component descriptions and specifications, instructions on how to use the graphical user interface to perform Ultrasound Guided Focused Ultrasound (USgFUS) therapy planning and delivery, guidelines on experimental data collection and data management capability.

408-GE Backshell Kit Assembly Instructions

This document provides assembly instructions for the backshell kit and guidelines for wiring element signals to pads on each of the cable termination boards.

Glossary of Special Terms Used by Verasonics

This document includes a comprehensive list of special terms used by Verasonics. It provides detailed descriptions and explanations for each term.

Probe Impedance Check User Guide

This document outlines the Probe Impedance Check tool, which is intended to provide users with a quick and simple way to measure the impedance of each element of a probe that is connected to the system. It is located in the “Tools” folder, at Tools / SimpleImpedance_Tool. It identifies any suspected shorts to ground (measured impedance less than about 10 Ohms) or open circuits (measured impedance greater than several hundred Ohms).

Vantage System 260-Pin ZIF Transducer Connector

This document provides information regarding the usage model of the ITT-Cannon 260-pin ZIF transducer interface connector used on the Vantage system.